

Extension: Perpendiculars and distance



- 1 Find the perpendicular distance d from the point (x_1, y_1) to the line $Ax + By + C = 0$.
- 2 Find the perpendicular distance from the origin to the line $4x + 7y + 5 = 0$.
- 3 Consider two parallel lines $9x - 9y - 9 = 0$ and $9x - 9y - 0 = 0$:
 - a Find the y -coordinate of the point on the line $9x - 9y - 9 = 0$ whose x -coordinate is 8.
 - b Use the point obtained from part(a) to find the shortest distance between the two given lines.
- 4 If the perpendicular distance from the point $(-5, -1)$ to the line $-2x - 7y + k = 0$ is $\frac{10}{\sqrt{53}}$, find the value(s) of k .
- 5 If the perpendicular distance from the point $(6, k)$ to the line $6x + (-2y) + (-6) = 0$ is $\frac{40}{\sqrt{40}}$, find the value(s) of k .
- 6 Consider the given point and the corresponding line:
Find the distance from the point to the line.
 - a $(3, 5), 4x + 3y + 1 = 0$
 - b $(4, -3), 24x + 7y - 3 = 0$
 - c $(5, 4), 5x - 12y = 0$
 - d $(-5, 3), 3x - 2y - 5 = 0$
- 7 Consider the points $A(5, 7), B(9, 10), C(6, 4)$ and $D(2, 1)$, which are the vertices of a parallelogram.
 - a Determine the equation of the line going through AB . Give your answer in the form $ax + by + c = 0$.
 - b If AB is the length of the parallelogram, find the perpendicular height of the parallelogram.
 - c Hence, find the exact area of the parallelogram.
- 8 Consider the lines $L_1: x + 2y + 5 = 0$ and $L_2: x + by + 20 = 0$.
 - a Find the distance between L_1 and the origin.
 - b Solve for the value(s) of b such that L_1 is twice as far from the origin as L_2 .